

machine learning. The flight efficiency software gives all the information in a terminal to the pilot, using AI algorithms based on previous flight data, present air traffic, weather data from satellites and sensors, as well as information from a large database of airports around the world helping them plot the best path for take-off and landing, which amounts for a large quantity of fuel consumption. The same data is also provided when the aircraft is airborne, with other inputs.

TAKE-OFF AND LANDING

The take-off and landing procedures are directly related to maintaining fuel efficiency onboard an aircraft. However, much of this technology is handled by three different suites of equipment - Flight Preview, Smart-Runway and SmartLanding. The latter of the two are part of the Enhanced Ground Proximity Warning System.

The Flight Preview system is loaded on an iPad, and has surprisingly lenient system requirements - an iPad 3 running on iOS 8.4 is good enough. It actually uses real-time satellite data to augment it on 3D-modelled surfaces, effectively giving pilots a clear view of the landing trajectory ahead. It costs region of about \$2 per iPad per month.

Chipping in here is the SmartPath Ground Based Augmentation System (GBAS). What this essentially does is use satellite data and radars to draw a "very precise, high integrity digital navigation data", which contributes to the technology available to air traffic control officials back on the ground. The key technology here, though, is the Enhanced Ground Proximity Warning System (EGPWS).

What we focus on here are the SmartRunway and SmartLanding systems - both being crucial elements of the latest Runway Awareness and Advisory Systems (RAAS) onboard aircrafts with digital instrumentation panels. The equipment is certified for flight instrumentation platforms that power both Airbus and

Boeing. These software elements take power from the onboard computers to facilitate active runway assistance for pilots. The systems run on the MFD of an aircraft with digital instrument



A typical, new-generation aircraft cockpit

clusters, i.e. most passenger aircrafts nowadays, and is key to using all the onboard imaging and sensory equipment for preventing accidents.

HANDLING WEATHER

Weather radars are key to flight safety. Take, for instance, the IntuVue RDR-4000 3D weather radar system, which comes with the added advantage of an easy-to-read interface - a key factor when you are reading emergency weather alerts. The radar provides a three-dimensional weather map of up to 180 degrees around the aircraft, ranging up to 320 nautical miles ahead of the aircraft and spanning all the way from ground to 60,000 feet. This helps pilots predict routes based on storms ahead, and request deviation from planned path well before hitting a storm cloud. It further provides 3D imagery of storm cells, and works in tandem with path-planning algorithms and the EGPWS to help alter routes and help during landings.

SURVEILLANCE SYSTEMS

Key improvement to the previously-established primary surveillance radar (PSR) and secondary surveillance radar (SSR) is the new Automatic Dependant Surveillance - Broadcast (ADS-B) radar, which improves surveillance, situational awareness, visibility and redundancy for safety majorly. The Airports Authority of India is in



Neelu Khatri, President of Aerospace at Honeywell India

line to upgrade aircrafts in its airspace to the new standards, in due time.

While all of this is good, this is one field where airlines seem to be lagging behind by a fair margin. The reasons include considerable need for investment, and an overhaul in existing surveillance infrastructure. Airspace technology, hence, is primarily sensitive and advanced in terms of using augmented reality, large databases, advanced algorithms and a host of redundancies in both hardware and software, which enable aviation to report sky-high figures of safety. Airline operators in India, too, are actively investing in the latest technologies to make flying more efficient, cost-effective, highly automated and even safer. The operators subscribing to these services also include India's largest volume air operator, IndiGo. Khatri says, "In 2015, local airline IndiGo signed a contract to feature their A320neo fleet with a suite of avionics technologies includ-



Tools like the EGPWS facilitate landing through any weather and condition, on almost any runway

ing safety focused cockpit systems that enable pilots to better monitor the surrounding airspace and provide greater situational awareness."

IN-FLIGHT CONNECTIVITY, AND THE WAY FORWARD

Khatri concludes a thorough investigation into the technology behind aviation, giving an insight into a peripheral yet critical-to-lifestyle change that is headed for Indian aviation sector, soon. Needless to say, aviation technology in India is in good hands as long as it keeps itself up to date with the latest advancements in navigation, surveillance, on ground and in flight technology. With the likes of Honeywell powering the same, the few shortcomings are likely to be met soon. **A**